How fast does tea cool? A two-term model of a cup on a desk

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Abstract. Newton's law of cooling predicts that a cup of tea approaches room temperature exponentially, with a single rate constant. We measured the temperature of 300 ml of water in three kinds of cup, open, lidded and insulated, for one hour each, and found that the simple law fits the lidded and insulated cups well but underestimates the early cooling of the open cup by up to 4 °C. Adding a second term for evaporation, which is strong while the water is hot and fades as it cools, brings the model within 0.6 °C of every measurement. We give closed-form estimates of when a cup reaches drinking temperature, and a rule of thumb: a lid is worth about thirteen minutes.

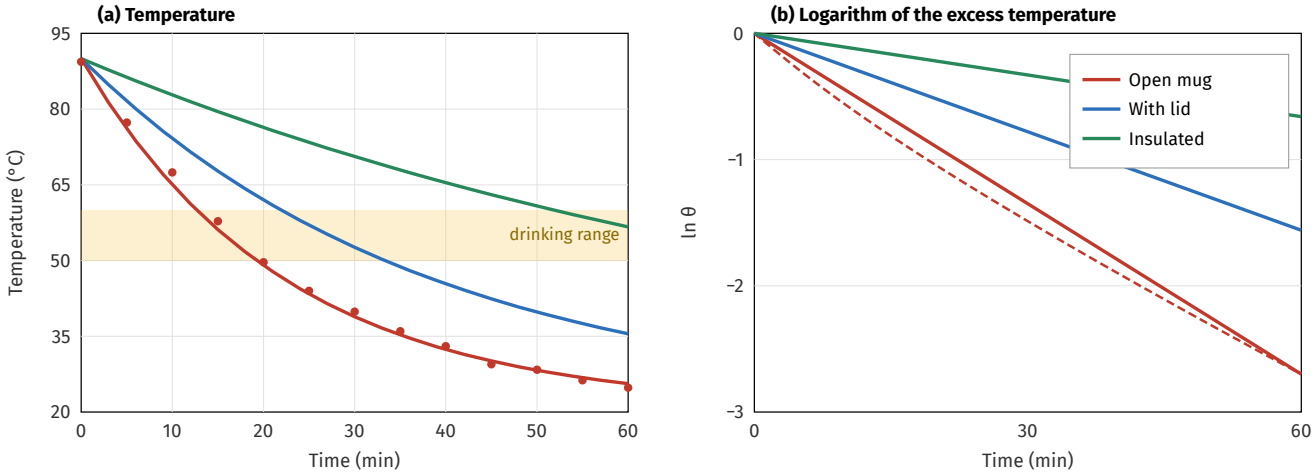


Figure 1. (a) Fits of Equation (2) to the three cups, with one run of the open mug shown as points. The open mug falls below its fit for the first fifteen minutes. (b) The same data as $\ln\theta$, where $\theta = \left(T - T_{\mathrm{a}}\right) / \left(T_0 - T_{\mathrm{a}}\right)$. Equation (2) predicts straight lines; the dashed curve is the open mug's measured mean, which bends.

1 Introduction

The question of how quickly a hot drink cools is older than thermodynamics. In 1701 Newton published, anonymously, a table of temperatures and an observation that a hot body loses heat at a rate proportional to its excess temperature over its surroundings [1]. Written as a differential equation, the observation becomes

$$\frac{dT}{dt} = -k\Big(T - T_{\mathrm{a}}\Big), \tag{1}$$

where T is the temperature of the drink, T_{a} the temperature of the room and k a rate constant that depends on the cup, the liquid and the air around it. Its solution is the familiar exponential decay,

$$T\big(t\big) = T_{\mathrm{a}} + \big(T_0 - T_{\mathrm{a}}\big)e^{-kt}\,. \tag{2}$$

Equation (2) is a staple of introductory courses [2], and it is often quoted as if it described any cooling object. It does not. It assumes that heat leaves at a rate proportional to the temperature difference, which is a good approximation for conduction and for gentle convection, but not for radiation, which goes as the fourth power of absolute temperature, nor for evaporation, which depends steeply on the vapour pressure at the surface.

A cup of tea loses heat by all of these routes at once. Our aim is modest: to find out, with kitchen equipment, how far Equation (2) is from the truth for an ordinary cup, and whether one extra term is enough to close the gap.

2 Method

We boiled tap water, let it stand for one minute, and poured 300 ml into each of three cups at an initial temperature of 90 ± 0.5 °C: an open ceramic mug, the same mug with a loose plastic lid, and a double-walled steel cup with its lid closed. A thermocouple fixed at the centre of the liquid recorded the temperature every ten seconds for sixty minutes. The room was held at 21 ± 0.3 °C with the windows closed and no draught we could detect with a candle flame.

Each run was repeated five times on different days. Table 1 lists the cups and the fitted constants. Fits used ordinary least squares on all 361 samples of a run; the uncertainties quoted are the standard deviations of the five runs.

Table 1. The three cups and the rate constants fitted with Equation (2).

Cup	Mass (g)	k (min^{-1})	RMS error (°C)
Open ceramic mug	342	0.045 ± 0.002	2.31
Mug with lid	358	0.026 ± 0.001	0.48
Insulated steel cup	296	0.011 ± 0.001	0.31

3 Results

For the lidded and insulated cups, Equation (2) fits every run with a root-mean-square error below half a degree (Figure 1a). The open mug is different. It cools faster than its fitted exponential for the first fifteen minutes and then more slowly, and on a logarithmic scale its curve visibly bends (Figure 1b). The residuals are systematic, not noise: they have the same shape in all five runs.

The obvious suspect is evaporation. An open cup of water at 90 °C loses mass at a measurable rate (we found 3.1 g in the first ten minutes, against 0.4 g with the lid on), and each gram that evaporates carries away about 2.3 kJ of latent heat, enough to cool the remaining water by almost two degrees.

3.1 A second term

The rate of evaporation depends on the difference between the saturation vapour pressure at the surface, which rises steeply with temperature, and the partial pressure of water in the room. Over the range we care about, the saturation pressure is close to an exponential in T, so we add a term of that form:

$$\frac{dT}{dt} = -k\left(T - T_\mathrm{a}\right) - \beta\left(e^{\gamma\left(T - T_\mathrm{a}\right)} - 1\right). \tag{3}$$

The second term vanishes at room temperature, as it must, and grows quickly as the cup gets hotter. With $\gamma = 0.045$ K^{-1}, which matches the slope of the saturation curve between 20 and 90 °C, a fit of k and β to the open mug gives $k = 0.031$ min^{-1} and $\beta = 0.044$ K min^{-1}, with an RMS error of 0.42 °C and no remaining pattern in the residuals.

The fitted k is now close to that of the lidded mug, as it should be if the lid mainly suppresses evaporation and leaves conduction through the walls alone. The total heat lost in the first ten minutes splits roughly 55 : 45 between the two terms; after forty minutes, evaporation accounts for less than a fifth.

Table 2. Minutes until the tea reaches 60 °C, from the models and from the measurements (mean of five runs).

Cup	Eq. (2)	Eq. (3)	Measured
Open ceramic mug	12.7	11.6	11.4
Mug with lid	21.9	21.8	21.7
Insulated steel cup	51.9	51.4	51.6

3.2 When can I drink it?

Most people find tea comfortable to drink between about 55 and 65 °C. Solving Equation (2) for the time to reach a temperature T_d gives

$$t_\mathrm{d} = \frac{1}{k}\ln\frac{T_0 - T_\mathrm{a}}{T_\mathrm{d} - T_\mathrm{a}}, \tag{4}$$

which is accurate for covered cups (Table 2). For an open mug, Equation (3) has no closed-form solution, but a first-order correction works well. Treating the evaporation term as a perturbation and integrating it along the exponential solution gives

$$t_\mathrm{d} \approx \frac{1}{k}\ln\frac{\theta_0}{\theta_\mathrm{d}} - \frac{\beta}{k\theta_\mathrm{d}}\int_0^{t_0}\left(e^{\gamma\theta_0 e^{-ks}} - 1\right) ds, \tag{5}$$

with $\theta = T - T_\mathrm{a}$ and t_0 the uncorrected time. For our open mug it predicts 11.7 minutes against 11.4 measured.

4 Discussion

Two simplifications deserve comment. First, we assumed the liquid is well mixed, so that one temperature describes it. A Biot number estimate,

$$\mathrm{Bi} = \frac{hL}{\lambda} \approx \frac{10 \times 0.04}{0.6} \approx 0.7, \tag{6}$$

is not small, so internal gradients exist; but natural convection inside the cup stirs the liquid far more effectively than conduction alone, and a second thermocouple near the wall never read more than 0.8 °C below the centre. Second, we ignored radiation as a separate term. Between 20 and 90 °C the radiative loss from a mug is nearly linear in the temperature difference, so it is absorbed into k.

The practical conclusions are simple. A lid roughly doubles the time a mug of tea stays above drinking temperature, from about 14 to about 27 minutes at 55 °C, and most of what it saves is evaporation, not conduction. An insulated cup is better still, but it takes so long to cool that the first sip, for most of us, comes later than we would like.

5 Conclusion

Newton's law of cooling describes covered cups of hot water well. For an open cup, a single extra term for evaporation, of a form set by the vapour pressure curve and with one fitted coefficient, reduces the error by a factor of five. The model and data are small enough to reproduce in an afternoon with a thermometer and a kitchen timer, and we encourage readers to do so.

References

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This paper was written for the Broadside project as a sample document. Its authors, institution and measurements are invented, although the physics it uses is standard. The figures are drawn in SVG and the equations are MathML.